

INFINITY ENERGY: The Future Of Self-Powered IoT

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Power remains the quiet obstacle slowing the global expansion of wearable and IoT devices. Despite rapid technological progress, these devices continue to depend on batteries that fail, leak, or require replacement, ultimately adding to e-waste within a few years. Thankfully, the idea of everyday devices such as TV remotes, fitness bands, smart switches, sensors, or even medical patches operating indefinitely without a battery is becoming increasingly feasible.

The principle itself is not new. Early wristwatches used kinetic energy to charge automatically, and crystal radios operated entirely without batteries. Modern energy-harvesting technology builds on these foundations, now capturing energy from multiple sources, including kinetic motion, vibrations, thermal gradients, and radio waves. This harvested energy can power or charge low-power devices, enabling them to operate for years without manual charging or even without a battery.

A previously published design demonstrates a batteryless energy-harvesting remote, available here:

<https://www.electronicsforu.com/electronics-projects/hardware-diy/remote-without-battery-that-can-work-for-years>

The same energy-harvesting principle underpins a system that can power low-power devices without a battery, or operate in combination with a supercapacitor and a battery to extend operating life. The system harvests energy from multiple sources, including thermal, solar, vibrational, and radio

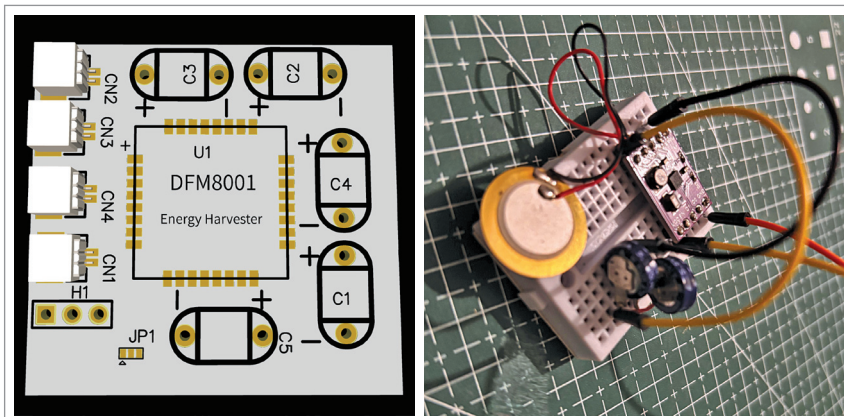


Fig. 1: DFM8001 energy harvesting system PCB (left) and LTC3588 energy harvesting system (right)

TABLE 1
BILL OF MATERIALS

Name	Designator	Quantity	Man. Part
10 μ F super capacitor	C1, C2, C3	1	NA
1 μ F super capacitor	C4, C5	2	NA
HC-1.0-2PWT	CN1, CN2, CN3, CN4	4	HC-1.0-2PWT
3-pin header VGT	H1	1	NA
HV0810-2R7105-R	SC1, SC2, SC3	3	HV0810-2R7105-R
.DFM8001	U1	1	DFM8001
LTC3588 module	U2	1	LTC3588
5.5V 47 μ F super capacitor	C6, C7, C8, C9	1	KR-5R5V474-R
Solar cell	PV1	1	
Piezoelectric	PZ1, PZ2, PZ3	3	
3.3V battery	BAT1	1	

waves, utilising almost any available ambient energy.

Two system designs are presented here, which are capable of powering devices either entirely without a battery or via a hybrid configuration that uses supercapacitor storage and battery backup for extended longevity. The systems intelligently combine harvested energy, supercapacitor storage, and battery support to supply power as available. The first design uses the DFM8001 IC module, and the

second design uses the LTC3588 energy-harvesting IC. Each offers distinct functions, features, and applications.

Fig. 1 shows the energy-harvesting system PCB based on the DFM8001 and the LTC3588 system. Components required to build this system are listed in the Bill of Materials in Table 1.

The two system designs explored here are:

1. The DFM8001 energy-harvesting power system